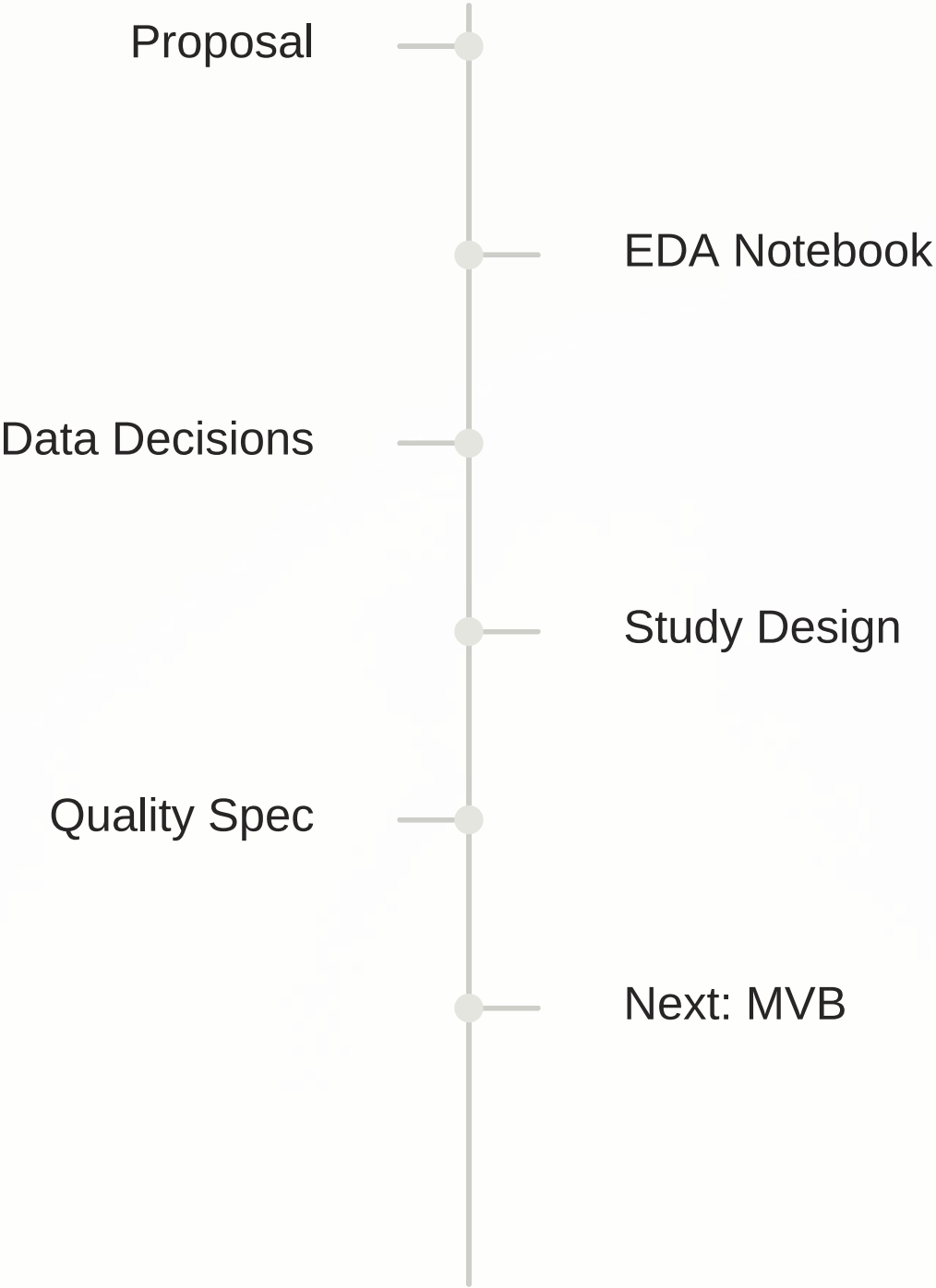


Sprint Update

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21/05/2026

From Raw GED Data to Pre-Registered Study

This week we moved from a broad proposal to a reproducible study setup: EDA complete, dataset locked, features revised, and study design formalised.



What Changed This Week

1

EDA Notebook

eda_ucdp_ukraine.ipynb

2

Locked Sample

data/sample_seed42.csv

3

EDA Outputs

data/eda_outputs/

4

Study Plan v0.2

study_plan.md

5

Study Design

study_design.md v0.1

6

Quality Spec

quality_spec.md v0.1

EDA Notebook: Reproducible Data Pipeline



Raw Data Not Pushed

Source CSV excluded from version control via
.gitignore

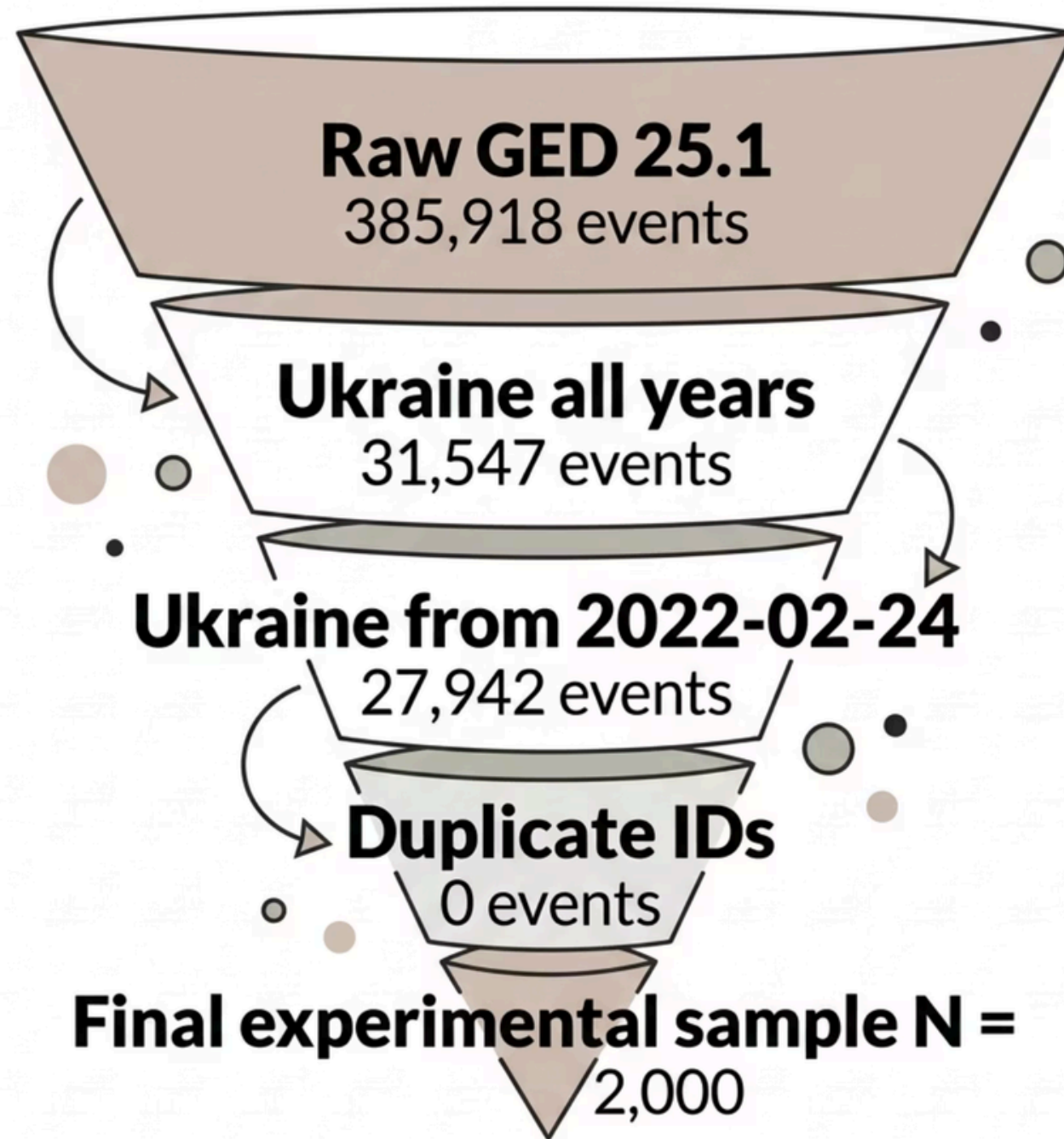
Derived Outputs Versioned

Sample and EDA outputs committed and
reproducible from seed

Dual Output

Notebook generates both visual plots and
data_provenance summary

Dataset Locked



Seed

42 — fixed for reproducibility

Date Range

2022-02-24 → 2024-12-31

Duplicate IDs

0 — no deduplication needed

Final Sample

N = 2,000 events

Feature Readiness: What EDA Told Us

EDA findings directly revised study_plan v0.2 — features were not assumed, they were evidenced.

✓ Keep

- latitude / longitude
- adm_1 oblast
- side_b actor
- best fatalities

✗ Drop

- type_of_violence — 99.2% state-based
- side_a — 100% constant
- adm_2 — 19.8% nulls, 128 categories
- where_description embeddings — low value

🔄 Transform

- best $\rightarrow \log(\text{best} + 1)$
- Reduces fatality skew for clustering

Clustering Risks Identified

99.2%

Event Type Imbalance

State-based violence only — near-zero variance in
type_of_violence

2

Fatality Skew

Median = 2, Max = 15,996 — log transform required

49.1%

Geographic Concentration

Donetsk oblast alone; top 3 oblasts account for 70.1% of events

1,518

Temporal Variation

Peak month: May 2024 — uneven temporal distribution across
sample

Geographic and Actor Structure

Coordinate Quality

- 100% valid coordinates — no missing geo data
- 97.1% geo-precision at levels 1–2
- side_a is constant: Government of Russia throughout
- side_b varies: Government of Ukraine and civilians



Design Limitation: Donetsk dominance (49.1%) means geographic clusters will be heavily biased toward a single oblast. This must be explicitly acknowledged in the study design and limitations section.

Geographic clustering is analytically defensible given the conflict theatre — but Donetsk concentration is a structural constraint, not noise.

Pre-Registered Study: $3 \times 2 \times 30 = 180$ Runs

Condition

F1: Location Only

F2: EDA-Informed Full

A — Baseline k-means

✓ 30 seeds

✓ 30 seeds

B — Human
conversational

✓ 30 seeds

✓ 30 seeds

C — LLM oracle

✓ 30 seeds

✓ 30 seeds

Design Controls

- K fixed across conditions for comparability
- Condition B blind to Silhouette score
- All outputs logged to JSONL
- 30 seeds per cell for robustness

Quality Spec + Next Steps

✓ Now Defined

Primary Quality

Silhouette score

Robustness

Davies-Bouldin index

Human Interpretability

Forced-choice pair rating

Oracle Validity

Spearman ρ + Cohen κ

Human Reliability

Krippendorff α

→
SOON

Next Sprint

01

Implement feature pipeline

02

Run Condition A baseline

03

Add configs and prompts

04

Write JSONL run logs

05

Confirm K via elbow method

06

Create human rating form + start oracle runs